

EnneadTab For Rhino

Shh... Secret Documentation
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BatchRenameBlocks

Tooltip: Block name batch editing utility for Rhino.

Features:

- Interactive table interface for block renaming
- Real-time block preview in viewport
- Double-click to isolate and inspect blocks
- Validates name conflicts automatically
- Preserves block definitions during renaming

Usage:

1. Edit desired names in the 'New Name' column
2. Double-click entries to preview blocks
3. Click 'Update Block Names' to apply changes

Access: Left Click

BlockDependencyMap

Tooltip: Report per-definition instance counts, embedded vs linked status, and the recursive nesting tree with max depth for every block definition in the document.

Helps answer "what uses this block?" and "how deep does this nesting go?" before editing or exploding a definition.

Access: Left Click



BlockFidelity

Tooltip: Classify every block definition in this document as embedded, linked, or proxy (linked but its source file is missing), and stamp the result into document user text.

Why this matters: Grasshopper definitions behave very differently against block-heavy models depending on whether their components round-trip blocks (names, materials, user strings, GUIDs preserved) or silently explode instances to raw geometry. This tool gives you the block side of that picture: which definitions are safe to edit here, and which are proxies pointing at missing files.

Features:

- Classifies all block definitions (embedded / linked / proxy)
- Shows a report with counts per class
- Stamps each block's fidelity into document user text (ET_BlockFidelity:) so the GH library catalog can read it

Rhino-side companion to the GH library's block-fidelity badge (feature 19).

Access: Left Click



EditDistortedBlock

Tooltip: Block editing utility for distorted instances.

Features:

- Creates editable copy of distorted block
- Maintains block definition relationships
- Automatic camera positioning for editing
- Temporary isolation of edited block
- Right-click to restore previous view

Access: Left Click



EditDistortedBlockRestoreView

Tooltip: Restore view after block editing.

Features:

- Restores previous camera position
- Shows all hidden objects
- Cleans up temporary editing blocks

Access: Right Click



FallGeosOnGeo

Tooltip: Project objects onto target geometry.

Features:

- Projects blocks using insertion points
- Projects other objects using bounding box centers
- Supports both top and bottom face projections
- Works with surfaces, polysurfaces and meshes
- Maintains object properties during projection

Access: Left Click



FilterBlockByName

Tooltip: Smart block selection utility.

Features:

- Filter blocks by name patterns
- Multi-select support for block names
- Sorted block name display
- Real-time search filtering
- Automatic selection of matching instances

Access: Left Click



IsolateSimilarBlocks

Tooltip: Isolate blocks of similar definitions from the selected blocks.

Features:

- Automatically identifies and isolates blocks with identical definitions
- Supports multiple block definitions at once
- Maintains original selection state
- Keeps other objects visible and unchanged

Access: Right Click



MakeBlockUnique / MBU

Tooltip: Create unique block definitions.

Features:

- Creates independent block definitions
- Optional name tagging with creator info
- Preserves block transformations
- Handles nested block structures
- Maintains layer assignments

Access: Left Click



MakeBlockUniqueToOne

Tooltip: Consolidate blocks into single definition.

Features:

- Merges multiple block types into one
- Creates new unified block definition
- Preserves instance positions
- Maintains transformation data

Access: Right Click



MatchTextureMappingInBlock

Tooltip: Copy texture mapping between blocks.

Features:

- Transfers texture mapping from source block
- Matches by layer correspondence
- Preserves mapping parameters
- Supports multiple target blocks
- Layer-specific texture application

Access: Left Click



PackageBlockLayer

Tooltip: Organize block content layers.

Features:

- Creates unified layer structure
- Optional layer flattening
- Preserves layer colors and materials
- Handles nested block hierarchies
- Streamlines material testing workflow

Access: Left Click



RandomBlocksOnSrfs

Tooltip: Advanced block distribution utility for surfaces.

Features:

- Distributes blocks randomly across target surfaces
- Configurable edge distance and spacing controls
- Optional edge-guided or curve-guided placement
- Real-time preview of block placement
- Supports multiple distribution patterns:
- Random interior placement
- Edge-aligned placement
- Even edge distribution

Usage:

1. Select target surfaces and sample blocks
2. Configure placement parameters
3. Choose distribution pattern
4. Preview and adjust as needed

Access: Left Click



RandomizeBlockTransformation

Tooltip: Randomly transform block transformation for rotation and scale.

Features:

- Rotates blocks randomly
- Scales blocks 1D height softly or taller
- Scales 3D dimensions evenly
- Animates transformation process

Access: Left Click



SampleLayout

Tooltip: Create sample block layout along crvs to quickly visualize design.

Features:

- Quick block layout visualization
- Flexible block size configuration
- Multiple layout modes:
 - Panel mode: Blocks span between divider points
 - Post mode: Blocks oriented to local coordinate of divider points
- User-friendly interface with clear step-by-step instructions
- Real-time preview of block placement
- Supports both open and closed curves
- Automatically handles curve segmentation for accurate block placement

Access: Left Click



SelectSimilarBlocks

Tooltip: Selects all block instances that share the same block definition as the selected blocks.

Usage:

1. Pre-select block instances (optional)
2. Run the command
3. Select additional blocks if none were pre-selected

Notes:

- Works with multiple block definitions at once
- Automatically filters for block objects only

Access: Left Click



SimplifyBlocks

Tooltip: Lighten a heavy model by reducing the mesh density inside block definitions.

Pick the blocks and set how much of the original mesh to keep with a slider. Each unique block definition is simplified once, and every instance of it in the model gets lighter at the same time.

Features:

- Slider control over how much detail is kept
- Reduction happens once per definition, not once per instance
- Placement, scale and rotation of every instance are preserved

Access: Left Click



ToggleBlockColorDisplay

Tooltip: Give every block type its own display color so you can tell them apart at a glance.

Useful when a model is full of near-identical block variations and you need to see which is which. Run it again to switch the coloring back off and restore the normal display.

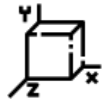
Access: Left Click



ToggleBlockColorDisplay_Setting

Tooltip: Toggle the on/off of block names.

Access: Right Click



UniformTransformGeos

Tooltip: Rotate or uniformly scale many objects at once, each about its own center.

Every selected block or object turns and resizes in place rather than orbiting a shared origin, so a street full of cars can all be spun or shrunk together without drifting out of position.

Usage:

1. Select the blocks or geometry to change
2. Enter the rotation angle, the scale factor, or both

Access: Left Click



QuickMassing

Tooltip: Create parametric massing models with advanced level editor interface.

Features:

- Interactive level editor table with reorder, add, remove functionality
- Dedicated surface picker with surface filter
- Real-time elevation calculations
- Persistent settings storage
- Visual level management interface

Usage:

1. Left-click to activate the QuickMassing tool
2. Use the level editor to configure building levels
3. Pick surfaces for massing creation
4. Generate massing model based on your specifications

Note: All settings are saved between sessions for convenience.

Access: Left Click



SlabEdge

Tooltip: Create custom slab edges by sweeping a profile block along selected edges.

Features:

- Multiple selection methods:
 - * Individual edge selection
 - * Edge loop selection
 - * Curve selection
- Profile block customization:
 - * Select from existing blocks
 - * Preview profile orientation
 - * Flip profile direction
- Interactive preview:
 - * Real-time visualization
 - * Adjust before finalizing
 - * Cancel and retry options

Usage:

1. Choose selection method
2. Select edges or curves
3. Pick profile block
4. Preview and adjust
5. Confirm to create

Note: Preview objects are automatically cleaned up after use.

Access: Left Click



StairMaker

Tooltip: Create parametric linear stairs with interactive controls and real-time preview.

Features:

- Dynamic stair creation with live preview
- Customizable parameters:
 - * Riser height
 - * Tread width
 - * Stair width
 - * Landing configuration
- Automatic step count calculation based on height
- Intelligent error handling for invalid inputs
- Support for multiple stair configurations
- Interactive point selection for precise placement

Usage:

1. Select start point for stair base
2. Define stair direction and height
3. Adjust width using flip point
4. Preview and confirm stair creation

Note: All parameters are validated in real-time to ensure code compliance.

Access: Left Click



StairMakerSpiral

Tooltip: Interactively create spiral stairs in Rhino.

Key Features:

- Dynamic spiral stair creation with preview
- Customizable riser height and tread width
- Adjustable spiral radius and rotation
- Automatic step count calculation
- Support for clockwise and counter-clockwise rotation

Access: Right Click



ColorPicker

Tooltip: Open an online color palette generator in your browser for scheme inspiration.

Starts you off on a preset EnneadTab palette that you can shuffle and lock until the mix feels right, then take the codes back into Rhino, Revit or a presentation.

Access: Left Click



DuplicateLayout

Tooltip: Duplicate layouts with customizable viewport offsets.

Key Features:

- Maintain layout settings and properties
- Adjust viewport positions with X-Y offsets
- Capture different model space views
- Preserve viewport scales and display settings
- Support for multiple layout selection

Access: Left Click



ExportSelectedLayout

Tooltip: Export selected layouts to PDF format.

Key Features:

- Multiple layout selection support
- Customizable output location
- Maintains print settings and quality
- Optional automatic PDF opening
- Batch processing capability

Access: Left Click



OpenSampleExcel

Tooltip: Access sample Excel template for area visualization.

Key Features:

- Pre-formatted Excel template
- Example data structure
- Color coding guidelines
- Area calculation formulas
- Category organization samples

Access: Right Click



PhoneScanner

Tooltip: Connect your phone camera to Rhino for instant photo & sketch scanning.

Features:

- Open pairing portal with QR code & PIN
- Real-time staging dock in Rhino for incoming photos
- One-click PictureFrame insertion into active Rhino document
- One-click copy image to system clipboard
- Batch insert multiple scans
- Automatic cleanup of ephemeral images

Access: Left Click



SectionCrowd

Tooltip: Create section view crowd representations.

Key Features:

- Interactive crowd placement in TOP view
- Customizable people spacing and density
- Random variations for natural appearance
- Support for multiple crowd patterns
- Automatic group creation for easy management

Access: Left Click



TagAnnotationObjects

Tooltip: Count and tag annotation objects (dimensions, hatches, text, leaders, text dots) in the active document.

Access: Left Click



VisualizeExcel

Tooltip: Convert Excel data into visual diagrams.

Key Features:

- Multiple shape options (circles, squares, bars)
- Customizable colors and sizes
- Automatic area calculations
- Support for grouped data visualization
- Dynamic layout adjustments.

Access: Left Click



ActivateEnneadTab

Tooltip: Restore EnneadTab functionality.

Key Features:

- Prefers git repo (developers) over EA_Dist (users)
- System path verification
- Component activation
- Path configuration
- Startup script setup

Access: Left Click



AppStore

Tooltip: Access EnneadTab's centralized tool repository.

Key Features:

- Complete tool collection
- Category organization
- Installation management
- Update notifications
- Tool documentation access

Access: Left Click



CheckErrorLog

Tooltip: Opens the Google error log URL in the default browser

Access: Left Click



ExtractPreviewImages

Tooltip: Extract preview images from Rhino files.

Key Features:

- Batch image extraction
- Multiple file support
- Automatic naming convention
- Progress tracking

Access: Left Click



GetEngine

Tooltip: Ensure that you have a localized Python engine installed

Access: Left Click



GetLatest

Tooltip: Update EnneadTab to latest version.

Key Features:

- Prefers git repo (developers) over EA_Dist (users)
- Automatic version detection
- Core module updates
- System path configuration
- Component synchronization
- Installation verification

Access: Left Click



HowToInstall

Tooltip: Access EnneadTab installation documentation.

Key Features:

- Step-by-step installation guide
- System requirements
- Troubleshooting tips
- Configuration instructions
- Team deployment guidance

Access: Left Click



MCPServer

Tooltip: Start the EnneadTab AI bridge for this Rhino session.

While it is running, an AI assistant can look at the open model and carry out changes for you in this session. Only the Rhino session you started it from is exposed, and the bridge stops when you close Rhino.

Access: Left Click



MakeANewButton

Tooltip: Create new EnneadTab button components.

Key Features:

- Interactive button creation
- Template generation
- File structure setup
- Icon integration support
- Documentation templates

Access: Left Click



OpenAutosaveFolder

Tooltip: Open the Rhino autosave folder locations.

Access: Left Click



OpenEcosystemFolder

Tooltip: Access the EnneadTab Ecosystem directory.

Key Features:

- Direct folder access
- System file management
- Resource exploration
- Configuration access
- Template management

Access: Left Click



ReloadEngine

Tooltip: Reload all EnneadTab modules to ensure latest changes are applied.

This script will:

1. Import all EnneadTab modules
2. Reload each module to apply latest changes
3. Handle any import errors gracefully

Access: Left Click



RemoteFixCode

Tooltip: Launches VS Code Dev for emergency remote code editing and debugging

Access: Left Click



ResetAllConduit

Tooltip: Reset all display conduits to default state.

Key Features:

- Complete conduit cleanup
- Memory optimization
- Display pipeline reset
- System performance improvement
- Debug assistance

Access: Left Click



RestartRhino

Tooltip: Restart Rhino to apply core updates.

Key Features:

- Safe application restart
- Update implementation
- Session state preservation
- Automatic core reloading
- System verification

Access: Left Click



SecretKeyBinding

Tooltip: Setup some secrete shortcut based on Sen's preference.

Access: Left Click



SelfRepair

Tooltip: Repair a stale EnneadTab installation and pull the latest version.

Run it when buttons are missing, point at the wrong tool, or still carry an old name. The toolbars are checked for out-of-date entries and, if any are found, the newest EnneadTab is fetched and installed for you.

Features:

- Spots toolbars left over from an older EnneadTab release
- Fetches and installs the current version automatically
- Nothing happens if your installation is already current

Access: Left Click



TellMeVersion

Tooltip: Display current EnneadTab Rhino version.

Key Features:

- Version number display
- Update status information
- Installation verification
- Component compatibility check
- Release notes access

Access: Left Click



Uninstall EnneadTab

Tooltip: Completely uninstall EnneadTab from the system

Access: Left Click



UnitTest

Tooltip: Execute unit tests for EnneadTab components.

Key Features:

- Comprehensive module testing
- Automated error detection
- System integrity verification
- Performance validation

Access: Left Click



YoutubePlaylist

Tooltip: Access EnneadTab's video tutorial library.

Opens the official EnneadTab YouTube playlist containing tutorials, demonstrations and workflow guides.

Access: Left Click



ArchiveFile

Tooltip: Archive files with customizable date prefixes.

Key Features:

- Multiple date format options
- Batch file processing
- Custom destination selection
- File opening options
- Automatic naming convention

Access: Left Click



AttachRefreshWorksession

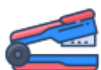
Tooltip: Attach worksession files or refresh the current worksession.

Key Features:

- Lists the files currently attached to the worksession
- Attach one or more .3dm / .rws files to the current worksession
- Refresh attached worksession files in one click
- Skips files that are already attached

Mirrors the widely-shared dharman Discourse snippets
(`'-Worksession Attach / Refresh'` via `rs.Command`).

Access: Left Click



BindWorksession

Tooltip: Convert worksession to single file with organized layers.

Key Features:

- Automatic layer organization
- File hierarchy preservation
- Progress tracking
- Resource cleanup
- Email notification support

Access: Left Click

CheckWorksessionFiles

Tooltip: Check an expected list of model files against the disk and the active worksession.

Reports which expected files are missing on disk, which are not attached to the worksession, and which attached models were not expected — then offers to attach the missing ones.

Access: Left Click



CreateWorksession

Tooltip: Create multi-file Rhino worksession.

Key Features:

- Multiple file selection support
- Custom session naming
- Automatic file linking
- Performance optimization
- Batch file processing

Access: Left Click



DocData

Tooltip: Demonstrates usage of RHINO_PROJ_DATA module for document data management.

This script showcases how to:

- Inspect existing document data
- Set preferred Grasshopper file path
- Store structured Grasshopper input parameters
- Handle various data types (bool, int, float)

Example Usage:

- Left-click to run the demonstration
- View results in Rhino command line

Access: Left Click



ExternalTrimmer

Tooltip: Manage external file trimming operations.

Key Features:

- External file linking and updating
- Boolean trimming operations
- Layer-based filtering
- Block instance management
- Automatic file synchronization

Access: Left Click



HarvestFolderReport

Tooltip: Scan a folder of .3dm files and report aggregated model context.

Key Features:

- Resolves a folder + file pattern (e.g. *.3dm), top folder or subfolders
- Lists every matched file before reading anything
- Aggregates layer paths found across all files
- Aggregates attribute user-text keys found across all files
- Prints a copy-paste-ready report to the command line

Rhino-side equivalent of the Rhino 8 Query Directory + Import Content
harvest pattern: batch-read model context without importing anything
into the active document.

Access: Left Click



ModelSnapshot

Tooltip: Versioned snapshots of model geometry, with Speckle-style branch/commit vocabulary, kept local.

Snapshot mode: capture the objects on chosen layers into a versioned store - a .3dm export plus a JSON manifest recording a fingerprint per object - on a named branch, with a commit message.

Diff mode: pick any past commit and compare it against the current document; added, removed, and modified objects are listed.

Manifests and exports live in /model_snapshots//.

Rhino-side companion to the GH library's model snapshot workflow (feature 18).

Access: Left Click



RebaseFile

Tooltip: Rebase file geometry and views to new origin point with X-axis orientation.

Key Features:

- Interactive base point and X-axis reference point selection
- Automatic object transformation with coordinate system reorientation
- View camera adjustment
- Named view preservation
- Comprehensive coordinate system update
- ETO form interface with clear instructions

Instructions:

- Every object, including those in locked and hidden layers, will be processed
- All views will be processed
- To avoid certain objects being rebased, put "REBASE_IGNORE" in the object's name
- To avoid certain cameras being rebased, put "REBASE_IGNORE" in the camera name

Access: Left Click



RefreshExternalLink

Tooltip: Update external linked references.

Key Features:

- External block updating
- Link synchronization
- Reference management
- File dependency tracking
- Automatic refresh handling

Access: Right Click



SaveSmallAndClose

Tooltip: Save optimized file and close document.

Key Features:

- Automatic resource cleanup
- Material purging
- Block definition optimization
- Layer cleanup
- Quick document closure

Access: Left Click



DVD

Tooltip: Classic DVD screensaver animation for Rhino.

A nostalgic entertainment feature that recreates the bouncing DVD logo animation within Rhino viewport.

Access: Left Click



EnneadCity

Tooltip: EnneadCity GUI Interface

Access: Right Click



PetDuck

Tooltip:

DuckiText - AI Architecture Assistant

A desktop companion that combines architectural expertise with AI capabilities.

Features:

- AI-powered chat and analysis
- Architecture tools and code library
- Rhino/Grasshopper integration
- Professional documentation support

Usage: Left click to activate

Access: Left Click



RedAlert

Tooltip: Command & Conquer inspired game mode for Rhino.

Features:

- Real-time audio feedback for modeling operations
- Classic RTS game sound effects
- Dynamic response to object creation/deletion
- Nostalgic gaming atmosphere while modeling

Access: Left Click



ChinaCodeRef

Tooltip: Open a China building code reference PDF without leaving Rhino.

Pick a document from the list and it opens straight away, or jump to the folder to drop in codes that are not there yet. The folder is created for you the first time you run it.

Usage:

1. Run the button and pick a code document from the list
2. Choose to open the folder instead if you want to add more PDFs

Access: Left Click



ET_LIB / LibrarySearch

Tooltip: Search the local library catalog with faceted chips.

Key Features:

- One search box: free text plus tag:/category:/author: facet chips
- Quoted values supported, e.g. tag:"panel unit"
- Unknown key:value tokens are kept as free text, never dropped
- author: chips match the sidecar author (the shared lib has no author facet)

Ports EnneadTab-For-Grasshopper#35 (faceted search).

Access: Left Click



ET_PAL / LibraryPalette

Tooltip: Keyboard-first quick launcher for the library catalog.

Key Features:

- One filter-as-you-type list combining pinned favorites and recently-opened definitions
- Favorites carry their one-keystroke shelf labels [1]..[9],[0] (GH feature 30)
- Enter opens the chosen definition in Grasshopper; Esc dismisses
- Opened entries are recorded at the front of the recent list

Ports EnneadTab-For-Grasshopper#60 (keyboard-first palette + favorites shelf).

Access: Left Click



EnneadTabGrasshopper

Tooltip: Launch the EnneadTab Grasshopper Library Explorer from inside Rhino.

Access: Left Click



ExpectedHost

Tooltip: Record which host document this Rhino model expects Grasshopper definitions to run against.

Definitions that drive the active Rhino document, a linked file (worksession attachment or model link), or an external host (Rhino.Inside.Revit) are not interchangeable. This tool stores the expectation in the document's user text (key ET_ExpectedHost) so the GH library explorer can hide definitions that don't match the current session.

Choices: activeRhinoDocument, linkedFile, externalHost, hostAgnostic.

Rhino-side companion to the GH library's host-model driver mode (feature 20).

Access: Left Click



GroupObjectsByKey

Tooltip: Group sample objects by a property key (Layer, Name, or user-text key) and preview the one-branch-per-value map.

Access: Left Click



HarvestModelContext

Tooltip: Inventory the active model's context for catalog tagging.

Features:

- Per-layer object counts with layer colors
- Block definitions with instance counts
- User-text key frequencies across objects
- Object type frequencies

Usage:

1. Run in a model
2. Copy the reported layer/block/user-text patterns into a GH catalog sidecar "modelContext" tag

Access: Left Click



LibraryAddress

Tooltip: Show a library entry's address stamp and check it for updates.

Pick a catalog entry to see its library address (entry id plus catalog version — the revision entry the Grasshopper importer stamps onto documents) and run update recognition: the catalog's recorded file hash is compared against the file on disk, flagging local changes with an asterisk, and an optional "version you have" answer produces the "v3 available — you have v2*" report. Ports the address stamp from the Grasshopper library explorer (EnneadTab-For-Grasshopper PR #46).

Reads the shared Grasshopper definition catalog through
Apps/_rhino/Library/catalog.py.

Access: Left Click



LibraryAiBadges

Tooltip: Show the AI enrichment badge for a catalog entry.

Key Features:

- AI description, tags, category and confidence from the sidecar
- Display only: never calls an AI service
- Flags low-confidence entries queued for human review

Access: Left Click



LibraryAssert

Tooltip: Run contract assertions for a catalog definition.

Key Features:

- Pick a Grasshopper definition from the shared library catalog
- Runs the author-declared red/green contract checks (PancakeContract-style)
- Uses live slider values when the definition is open in Grasshopper, else catalog defaults
- Warns before opening a definition whose checks fail

Access: Left Click



LibraryBackup

Tooltip: Back up the catalog's sidecar/metadata files to a timestamped zip on demand.

Key Features:

- Snapshots every *.ennead.json sidecar under each catalog scan root
- Stores a backup-interval preference in library-settings.json (backupIntervalMinutes)
- Offers 'back up now' when the newest backup is older than the interval
- Prunes old backups, keeps the newest 10

Ports EnneadTab-For-Grasshopper#57 (interval auto-backup of sidecars/metadata).

Access: Left Click



LibraryCanvasPreview

Tooltip: Show a static summary of a Grasshopper definition's canvas without opening Grasshopper.

Rhino port of the static canvas preview (EnneadTab-For-Grasshopper PR #33): a .ghx file is plain XML, so its canvas is summarized directly -- component type counts and names, group and scribble counts, document id, and canvas bounds -- plus the plugin assembles the catalog index knows about. Binary .gh files cannot be parsed this way and are skipped with a note. Read-only; nothing is opened or modified.

Access: Left Click



LibraryContract

Tooltip: Show a library definition's Hops-style input/output contract.

Pick a catalog entry to see the contract the Grasshopper plugin derived at index time: inputs grouped by control kind (sliders, toggles, value lists, panels, referenced geometry, Get components) with type, default, domain, and required/optional flags, followed by terminal outputs. Ports the auto-derived parameter contract from the Grasshopper library explorer (EnneadTab-For-Grasshopper PR #37).

The live Grasshopper derivation needs a running Grasshopper document, so from Rhino this button shows the contract stored in the catalog index; entries indexed before contract derivation report that none is stored.

Access: Left Click



LibraryCopy

Tooltip: Copy a catalog entry's path or JSON to the clipboard.

Key Features:

- Pick an entry from the library catalog
- Copy Path: the entry's source path
- Copy JSON: the entry's full JSON (id, title, description, path, tags, category, parameters, sidecar)

Ports EnneadTab-For-Grasshopper#19 (explorer copy actions).

Access: Left Click



LibraryDryRun

Tooltip: Preview a catalog refresh without writing anything.

Key Features:

- Reports what a refresh would do: reuse / reindex / drop / error previews
- Scans the configured roots read-only (no Rhino open, no index write)
- Flags missing or unreadable roots before a real refresh hits them

Access: Left Click



LibraryDupes

Tooltip: Find duplicate file content across catalog entries and report the groups.

Rhino port of the Grasshopper library duplicate-content report (EnneadTab-For-Grasshopper PR #27): entries whose files are byte-identical but stored at different paths are grouped by content hash so the copies can be cleaned up.

Key Features:

- SHA-256 content hash per entry file (no mtime/size fingerprints)
- Same-path repeats are collapsed, never reported as duplicates
- Groups ordered by hash; entries within a group ordered by path
- Pick a group to see every copy's path; full report on the command line

Access: Left Click



LibraryExamine

Tooltip: Read-only detail view of one catalog entry -- opens nothing, changes nothing.

Pick a Grasshopper library catalog entry to see a read-only summary: title, category, tags, parameter contract summary, provenance (scan root, index time, file hash, convention check), and the purpose note from its sidecar.

Ports the read-only "Examine" preview idea from EnneadTab-For-Grasshopper PR #48. On the Rhino side the preview is a single detail dialog instead of a locked Grasshopper canvas: a definition is never opened and nothing is modified. Depends on PR #283 (shared catalog lib); degrades gracefully if unmerged.

Access: Left Click



LibraryExport

Tooltip: Export the filtered library catalog to a portable JSON file.

Key Features:

- Optional filter: free text plus tag:/category:/author: facet chips
- Portable shape: id, title, sourcePath, tags, category, parameters
- Save dialog; creates parent folders; UTF-8 JSON

Ports EnneadTab-For-Grasshopper#16 (catalog export).

Access: Left Click



LibraryFavorites

Tooltip: Pin and unpin catalog definitions as favorites.

Key Features:

- Pin entries so favorites sort first in pin order
- Pin order is preserved across sessions
- Stored in the shared library-settings.json (favoriteEntryIds)

Access: Left Click



LibraryInputSummary

Tooltip: Summarize a library definition's inputs, grouped by control type.

Pick a catalog entry to see its Inputs-group parameters grouped as sliders, toggles, value lists, panels, referenced geometry, or other, with counts and member names — the "drive 3 sliders" view from the Grasshopper library explorer (EnneadTab-For-Grasshopper PR #42). The picker list itself shows the compact summary (e.g. "2 sliders · 1 toggle") next to each entry.

Reads the shared Grasshopper definition catalog through
Apps/_rhino/Library/catalog.py; inputs come from the catalog index.

Access: Left Click



LibraryInsertModes

Tooltip: Open a catalog definition in Grasshopper -- three open modes.

Ports EnneadTab-For-Grasshopper PR #53 (insert as loose / group / cluster).
The Rhino-side equivalent of inserting into a canvas is opening the definition in Grasshopper:

- Loose: open the .gh normally in Grasshopper.
- Grouped with model: copy the .gh into the current .3dm's folder, then open the copy (keeps the definition next to the model it belongs to).
- Pinned version: open a chosen pinned/versioned copy of the definition (sidecar "pinnedVersion", or versioned sibling files such as Foo.v2.gh).

Invocation: the Grasshopper plugin's explorer is a canvas component and exposes no Rhino command entry point, so this button opens documents through Grasshopper.Instances.DocumentServer.AddDocument(path, True) from Rhino IronPython, loading the Grasshopper plug-in first with the _Grasshopper command when it is not already loaded.

Depends on PR #283 (shared catalog lib); degrades gracefully if unmerged.

Access: Left Click



LibraryInsights

Tooltip: Show a read-only Insights dashboard for the Grasshopper definition catalog.

Rhino port of the Grasshopper catalog analytics dashboard (EnneadTab-For-Grasshopper PR #49): entry counts per category and tag, most-used plugins, sidecar completeness (how many entries have a *.ennead.json and how many are clean), and -- when a library-usage.json file is present -- a usage section with the most-opened entries and the stale-entry cleanup hit-list (no opens in 90 days). Pure report; nothing is modified.

Access: Left Click



LibraryLint

Tooltip: Validate every catalog sidecar and report unknown or mistyped fields.

Rhino port of the Grasshopper library sidecar field warnings (EnneadTab-For-Grasshopper PR #23): sidecars whose fields have the wrong type or are unknown no longer fail silently - each problem is collected as a structured warning so typos (e.g. 'catagory') surface instead of being dropped.

Key Features:

- Checks all catalog entries through the shared Library catalog module
- Reports unknown sidecar fields
- Reports mistyped values (wrong JSON type for a known field)
- Full per-entry report printed to the command line

Access: Left Click



LibraryOpenWithInputs

Tooltip: Open a catalog definition with its annotated inputs.

Key Features:

- Pick a Grasshopper definition from the shared library catalog
- Fill Player-style annotated inputs (min/max clamp, presets) in a Rhino form
- Opens the definition in Grasshopper and applies the values to matching sliders

Access: Left Click



LibraryPackageRestore

Tooltip: Show the Yak install commands for a catalog entry's missing plugins (ports EnneadTab-For-Grasshopper PR #74).

Reads the entry's required plugin packages ("DisplayName|secret" wire format, same as the Grasshopper plugin), compares them against the plugins installed on this machine, and emits the `yak install` commands needed to restore the missing ones.

Read-only helper: it never runs installs itself. Copy the commands to the clipboard or open the Rhino PackageManager from the prompt.

Key Features:

- Missing vs already-installed plugin report per entry
- Install hints prefer the secret GUID (unambiguous on the package server)
- Copyable command list (clipboard + command line)
- One-click open of the Rhino PackageManager

Access: Left Click



LibraryParamPreview

Tooltip: Show a library definition's parameter contract, grouped by parameter group.

Pick a catalog entry to see its parameter summary header (total plus per-group counts) followed by one name/type/default table per group, in first-seen group order. Ports the grouped parameter preview from the Grasshopper library explorer (EnneadTab-For-Grasshopper PR #15).

Reads the shared Grasshopper definition catalog through Apps/_rhino/Library/catalog.py; parameters come from the catalog index.

Access: Left Click



LibraryPinnedImport

Tooltip: Import a library definition at a pinned version.

Key Features:

- Pick a catalog entry and one pinned version from its sidecar history ("versions": [{version, diffNote, publishedAt, sourcePath, fileHash}, ...])
- Copy that version's .gh file to a user-chosen folder (file name stamped with the version), or open it in Grasshopper
- Rhino-side equivalent of the GH pinned import (cluster/Hops/snippet): the copy is a frozen snapshot -- library updates never change it

Ports EnneadTab-For-Grasshopper#52 (pinned-version import as cluster/Hops/snippet).

Access: Left Click



LibraryPopular

Tooltip: Catalog definitions ranked by usage.

Key Features:

- Reads the Grasshopper usage store (library-usage.json)
- Ranked by opens, then solves
- Display only: never writes usage data

Access: Left Click



LibraryPortability

Tooltip: Audit a catalog definition for portability problems (ports EnneadTab-For-Grasshopper PR #72).

Checks one library entry for the things that break when a definition moves to another machine: absolute paths hiding in the sidecar or in indexed parameter values, plugin assemblies the entry needs but this machine does not have installed ("Name|secret" wire format, cross-checked against the Grasshopper Libraries / yak package folders and the entry's pluginsKnown provenance flag), and Windows-only assumptions (registry keys, drive-letter paths, Program Files references).

Key Features:

- Pass / warn / fail verdict per entry with fix hints
- Absolute-path scan over sidecar fields and parameter values
- Missing-plugin detection with yak install hints
- Full report printed to the command line

Access: Left Click



LibraryProvenance

Tooltip: Show provenance for a catalog entry: will this definition open on my machine?

Rhino port of the Grasshopper provenance tab (EnneadTab-For-Grasshopper PR #51): component count and component kinds (parsed from the .ghx XML when available), the plugin assemblies the definition needs (from the catalog index), and a best-effort check flagging required plugins that are not installed (scans loaded assemblies plus the Grasshopper Libraries folders for .gha files). Read-only; nothing is modified.

Access: Left Click



LibraryPublish

Tooltip: Stage a Grasshopper catalog entry for publishing (ports EnneadTab-For-Grasshopper PR #69).

Picks a library catalog entry (or browses for a .gh/.ghx file), collects the publish metadata (title, version, description, tags, category), validates the request the same way the Grasshopper publish dialogs do, writes the versioned *.ennead.json sidecar, then stages the definition + sidecar (+README if one sits next to the definition) into a publish/staging folder under the catalog directory.

Key Features:

- Entry picker over the shared catalog index, or browse for a file
- Next-version suggestion ("1.2.0" -> "1.2.1", "v3" -> "v4", "3beta" -> "4beta")
- Request validation (source file exists, title, dotted-numeric version, no blank tags)
- Sidecar field warnings through catalog.validate_sidecar
- Full staging report printed to the command line

Access: Left Click



LibraryPublishers

Tooltip: List named publishers harvested from catalog sidecars and rename one everywhere.

Catalog entries can expose contract parameters as named publishers (the sidecar "publishers" map: publisher name -> contract parameter name, borrowed from Telepathy by EnneadTab-For-Grasshopper PR #47). Pick a publisher, enter a new name, and the rename propagates across every sidecar that references it, the version-agnostic markdown docs (.gh.ennead.md), and any saved contract presets (contract-presets.json) -- one rename, everywhere consistent.

Additive-only: sidecar files keep every existing field; only the publisher key is renamed. Depends on PR #283 (shared catalog lib); degrades gracefully if unmerged.

Access: Left Click



LibraryPurposeNotes

Tooltip: View and edit purpose notes for library definitions, keyed by document GUID.

Key Features:

- Pick a catalog entry and view its stored purpose note
- Edit the note; it is saved back into the entry's .enneed.json sidecar under an additive "purposeNotes" key, keyed by document GUID ("doc:" when the definition file is readable, "file:" fallback) -- existing sidecar keys are never renamed or touched
- Note schema mirrors the GH definition-notes.json format:
{"purposeNotes": {"doc": {"text": "...", "updatedAt": "..."}}}

Ports EnneedTab-For-Grasshopper#55 (purpose notes keyed by document GUID).

Access: Left Click



LibraryReadme

Tooltip: Show a library definition's version-agnostic README.md docs.

Pick a catalog entry to read the .gh.README.md sidecar sitting next to the definition file — usage notes, input explanations, and known limitations, exactly as the Grasshopper library explorer renders them (EnneedTab-For-Grasshopper PR #30). The docs live outside the .gh file, so updating the definition never wipes them; files over 256 KiB are not loaded.

Reads the shared Grasshopper definition catalog through Apps/_rhino/Library/catalog.py to resolve each entry's definition path.

Access: Left Click



LibraryRecent

Tooltip: Recently-opened catalog definitions (MRU list).

Key Features:

- Newest-first list of opened definitions, capped at 10
- Opening an entry re-records it at the front
- Stored in the shared library-settings.json (recentEntryIds)

Access: Left Click



LibraryRefresh

Tooltip: Refresh the Grasshopper definition catalog (throttled).

Key Features:

- Re-scans the configured roots and rebuilds the catalog index
- Throttled: skips the re-scan while the last refresh is still fresh
- Interval is configurable in the library settings; 0 disables throttling

Access: Left Click



LibraryScanRoots

Tooltip: Validate the catalog scan roots.

Key Features:

- Checks every configured scan root for missing or unreadable paths
- Warn-and-continue: reports all problems at once, flags possibly-offline shares

Access: Left Click



LibrarySort

Tooltip: Sort the local library catalog by title, category, or path.

Key Features:

- Ascending / descending sort on title, category, or path
- Remembers the last sort choice in the library settings
- Shows the sorted catalog in a pick list

Ports EnneadTab-For-Grasshopper#13 (SortBy/SortDirection).

Access: Left Click



LibraryVersions

Tooltip: Version history for a library definition: update check + roll back.

Key Features:

- Pick a catalog entry and list its version history from the .enneed.json sidecar ("versions": [{version, diffNote, publishedAt, sourcePath, fileHash}, ...]); the live version is marked [current]
- Roll back: restores a versioned backup copy of the .gh + its sidecar over the live files; the live files are first backed up to .bak copies, and the button asks for confirmation before overwriting anything
- Update check: compares the live file's last-modified time against the latest version's publishedAt / recorded file and reports whether it looks current

Ports EnneedTab-For-Grasshopper#50 (version history + Update / Roll back).

Access: Left Click



LibraryViews

Tooltip: Saved named filtered views of the local library catalog.

Key Features:

- Save the current search (free text + tag:/category:/author: chips) as a named view (author: chips match the sidecar author; the shared lib has no author facet)
- Apply a saved view to filter the catalog
- Delete saved views; views persist in the library settings

Ports EnneedTab-For-Grasshopper#32 (bookmarkable filtered views).

Access: Left Click



LibraryWhy

Tooltip: Explain why a catalog search hit matches.

Key Features:

- Shows which fields and terms matched each hit
- One line per hit, e.g. matches via tag 'nesting' and title 'panel'
- Mirrors the Grasshopper match-explanation wording

Access: Left Click



LibraryYakRestore

Tooltip: Install missing Grasshopper plugins for a library entry via Yak. Reads the entry's required plugins ("Name|secret" format), detects installed plugins from the Grasshopper Libraries folder and Rhino Yak package folders, offers to run `yak install` for the missing ones and reports results.

Ports EnneadTab-For-Grasshopper#74 (missing-plugin package restore via Yak).

Access: Left Click



ModelContract

Tooltip: Edit the living model-contract description for a model context key, and propagate it to every Grasshopper catalog sidecar referencing that context.

A model contract describes what a definition expects from the model (for example which layers and user-text keys it reads). Catalog entries carry a modelContextKey; this tool edits the shared description once and writes it into every *.gh.ennead.json / *.ghx.ennead.json sidecar in your library folder whose modelContextKey matches, and stores it in this document's user text too.

Features:

- Pick an existing context key or create a new one
- Multiline editor for the contract description
- Stores the description in the document (user text key ET_ModelContract:)
- Propagates to every matching catalog sidecar in a library folder
- Remembers the library folder between sessions

Rhino-side companion to the GH library's model-contract propagation (feature 17).

Access: Left Click



ModelFootprint

Tooltip: Live footprint of layer patterns against the current model.

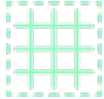
Features:

- One or more layer wildcard patterns, comma-separated (e.g. FACADE::*, *_Annotation)
- Per-pattern live object counts with layer color swatches (hex)
- Remembers the last patterns used

Usage:

1. Enter layer wildcard patterns
2. See the live match count per pattern

Access: Left Click



PerforationRatio

Tooltip: Find out how to calculate your perforation panel with precise opening ratio.

Access: Left Click



PlaceAsset

Tooltip: Place Asset from asset library

Access: Left Click



SearchCommand / LearnEnneadTabForRhino / CommandList

Tooltip: Search and learn EnneadTab commands.

Key Features:

- Interactive command search
- Function documentation
- Command aliases
- Tool location finder
- Visual command preview

Access: Left Click



SlopeCalculator

Tooltip: Open an online slope calculator in your browser.

Converts between slope angle, percentage and ratio, which is the conversion you want when checking a ramp against accessibility limits or working out a road or roof pitch.

Access: Left Click



Tutorial

Tooltip: EnneadTab learning resources hub.

Features:

- Access to comprehensive GH tutorials
- Local documentation and PDF guides
- Video tutorials via YouTube playlist
- Quick reference materials for common workflows

Access: Left Click



Anything

Tooltip: Run the EnneadTab Lab scratch script, a sandbox for trying out ideas in Rhino.

Kept as a testing ground for behavior that is still being explored, so it does not read or change anything in your model. Whatever it does is reported on the Rhino command line.

Access: Left Click



Dockpane

Tooltip: A dockable panel that attaches to the side of Rhino window

Access: Left Click



Text2Script

Tooltip: Utility script to convert text to script using AI.

Note: This is Rhino 8 only.

Features:

- Converts natural language to executable Python script
- Integrated with Rhino's rhinoscriptsyntax
- Maximum 5 refinement attempts for optimal results
- Always uses main() as the entry function
- Modern error handling and user feedback

Access: Left Click



Text2ScriptSetting

Tooltip: Report the current settings of the Text2Script generator on the Rhino command line.

Right-click companion to Text2Script, the tool that turns a plain-English request into a runnable Rhino script. Nothing in your model is read or changed.

Access: Right Click



ViewPrettifier

Tooltip: Generate AI renderings from architectural images.

Features:

- Processes existing architectural images
- Detects edges in the image to preserve model structure
- Allows use of style reference images
- Modern dark-themed GUI for configuration
- Model and ControlNet selection
- Progress tracking with step indicators
- Generates AI renderings using Stable Diffusion with ControlNet
- Shows real-time progress during generation
- Automatically installs required dependencies
- Monitors for stalled processes and provides diagnostics

Access: Left Click



ApplyColorBook

Tooltip: Apply the office Color Book to Rhino layer colors - no Excel needed.

Pulls the project's resolved color book from enneadtab.com (Department + Program swatches) and recolors every layer whose name or abbreviation matches a swatch. Sign in once when prompted.

Key Features:

- Online source: colors live at enneadtab.com/color-book, not a scattered Excel
- Matches layers by name OR abbreviation (case-insensitive, exact)
- Reports how many layers were recolored
- Same office standard the Revit ColorScheme tool uses (shared COLOR library)

Access: Left Click



DestroyLayer

Tooltip: Delete layers outright, even when they still hold objects.

Rhino normally refuses to remove a layer that is in use, especially when a block definition is holding it hostage. This clears the objects out and takes the layer with them, so a bloated layer tree can finally be pruned.

Usage:

1. Run the button and pick the layers to remove
2. The layers and everything on them are gone once you confirm

Access: Left Click



FindLayerInFiles

Tooltip: Search for layers across multiple files.

Key Features:

- Multi-file layer search
- Case-insensitive matching
- Progress tracking
- Search result summary
- File origin tracking

Access: Left Click



ForceLayerPackage

Tooltip: Detach selected geometry and blocks to a chosen SYSTEM_ layer.

Workflow:

1. Detect or create SYSTEM_XX root layers.
2. Ask user to choose the target SYSTEM_ layer through a list-box.
3. Migrate non-block geometry first, preserving any sub-layer structure.
4. Make each block definition in the selection unique so that edits do not affect instances outside the selection.
5. Update every object (including objects inside the new block definitions) so that their layers live under the chosen SYSTEM_ layer.

The implementation is written as a group of small helper functions for future reuse.

Access: Left Click



InitiateLayers

Tooltip: Create standardized layer structures.

Key Features:

- Predefined layer schemes
- Color-coded organization
- Program-based grouping
- Material-based grouping
- Customizable hierarchy

Access: Left Click



IsolateLayerBySelection

Tooltip: Show only objects on selected layers.

Key Features:

- Selection-based isolation
- Layer visibility control
- Quick view focusing
- Multiple layer support
- View organization

Access: Left Click



LayerNameFormat

Tooltip: Standardize layer naming conventions.

Key Features:

- Multiple format options
- Case formatting
- Pattern preservation
- Batch renaming
- Protected name handling

Access: Left Click



LayerStateSnapshot

Tooltip: Snapshot and restore layer visibility states (session-scoped named snapshots).

Access: Left Click



MergeLayer

Tooltip: Combine multiple layers into one.

Key Features:

- Layer consolidation
- Block-aware merging
- Property preservation
- Layer cleanup
- Batch processing

Access: Left Click



NestLayer

Tooltip: Organize layers in hierarchical structure.

Key Features:

- Layer nesting automation
- Block-aware processing
- Duplicate name handling
- Layer hierarchy preservation
- Batch layer organization

Access: Left Click



RandomLayerColor

Tooltip: Assign random colors to layers.

Key Features:

- Smart color assignment
- Context-aware coloring
- Desaturated color option
- Color scheme preservation
- Batch processing support

Access: Left Click



RandomLayerColorSetting

Tooltip: Change the setting of color style.

Access: Right Click



SelectObjectsOnSimilarLayer

Tooltip: Selection objects on the similar layers.

Access: Right Click

ValidateLayerPatterns

Tooltip: Validate Elefront-style layer patterns (with :: full paths and * wildcards) against the document's layer table.

Reports per-pattern match counts and, for patterns that match nothing, suggests the closest real layer names — catching renamed layers before a definition fails silently.

Access: Left Click



AssignEmptyMaterialToLayer

Tooltip: Same as EA_AssignEmptyMaterial

Access: Left Click



ImportSelectedMaterial

Tooltip: Import materials from external files.

Key Features:

- Selective material import
- Property preservation
- Material preview
- Name conflict handling
- Multi-select support

Access: Left Click



LayerMaterialSwatchReport

Tooltip: Per-layer material swatch report with optional assignment.

Key Features:

- Lists every layer with its material name
- Shows diffuse color hex per material
- Flags layers falling back to the default material
- Optional one-click material assignment to layers

Access: Left Click



MaterialPrefix

Tooltip: Add file-specific prefixes to materials.

Key Features:

- Automatic prefix generation
- Session compatibility
- Name conflict resolution
- Batch processing
- Material tracking

Access: Left Click



MergeMaterials

Tooltip: Consolidate multiple materials into one.

Key Features:

- Material consolidation
- Block-aware processing
- Layer material handling
- Object material updating
- Automatic cleanup.

Access: Left Click



RandomTextureWalk

Tooltip: Randomize texture mapping positions.

Key Features:

- Texture offset randomization
- Mapping preservation
- Customizable range
- Batch processing
- Pattern variation

Access: Left Click



RemoveStringInMaterialName

Tooltip: Remove the specific string in material name. Handy if trying to remove file name prefix.

Access: Left Click



AuditUserTextKeys

Tooltip: Audit user-text keys on the selected objects (or the whole document) and report per-key object counts.

Shows how many objects carry each key plus the number of distinct values, so missing, misspelled, or inconsistently filled keys surface before baking.

Access: Left Click



BakeNameClean

Tooltip: Delete previous bake outputs by bake-name pattern (idempotent bakes).

Features:

- Matches object names and/or the "BakeName" user-text key
- Wildcard patterns (e.g. BriseSoleil_*)
- Preview count with explicit Yes/No confirmation before deleting

Usage:

1. Enter a bake-name wildcard pattern
2. Choose the match source (name, BakeName user-text, or both)
3. Confirm the deletion

Access: Left Click



BakeWithAttributes

Tooltip: Bake selected objects with an explicit name / layer / color mapping.

Key Features:

- One dialog for the full attribute mapping, no hidden defaults
- Name template with {} index token (e.g. 'Panel_{i}')
- Layer path with '::' nesting, created on demand
- Display color applied per object (color source = object)
- Copies the selection; originals are untouched

Rhino-side equivalent of the Grasshopper ModelObject -> ObjectAttributes translation fallback: name, layer path and display color are documented in the contract before anything is baked.

Access: Left Click



CleanStaleUserText

Tooltip: Audit and clean stale attribute user-text keys in the active model.

Key Features:

- Audits object + document user-text keys before changing anything
- Flags stale 'Grasshopper Bake Data' leftovers (known file-bloat cause)
- Flags empty-valued orphaned keys no definition reads anymore
- Checklist UI: you pick exactly which keys to remove
- Reports how many objects each key was removed from

Mirrors the community 'Grasshopper Bake Data' hygiene scripts and Elefront's Remove User Text.

Access: Left Click



ExplicitBake

Tooltip: Bake with an explicit Push/Pull/Bake/Purge action mode and a required unique base name.

Access: Left Click



FlattenMeshFace

Tooltip: Try to flatten the mesh face so there is no bump

Access: Left Click



MakeVoidSeam

Tooltip: Creates void cut polysurfaces based on layer name configuration.

The script processes layers containing seam definitions to generate corresponding void cuts. Layer names should include width parameters in square brackets (e.g. 'seam[12]').

Access: Left Click



MatchCrvDir

Tooltip: Match multiple crvs direction.

Access: Left Click



Measure3D

Tooltip: Display 3D measurements between two points.

Shows total distance with a curve, and displays X, Y, Z differences as color-coded curves:

- X difference in red
- Y difference in green
- Z difference in blue

Measurements remain on screen until a new measurement is started.

Access: Left Click



MoveFixedDist

Tooltip: Move selected objects a fixed distance in 6 directions.

Features:

- Simple modeless interface with 6 directional buttons
- Persistent distance value between sessions
- Input validation for numeric values
- North/South/East/West/Up/Down movement
- Allows full interaction with Rhino while dialog is open
- Gamepad-style button layout
- Zoom to selected function
- Measure3D function for quick measurements
- Choice between View CPlane and World CPlane coordinate systems

Access: Left Click



OffsetFloorBorder

Tooltip: Shrink or expand the outline of a floor plate by the distance you type.

Feed it flat surfaces or solid slabs and the edge moves in or out, which is how you rough out a setback, a balcony line or a shaved core without redrawing the boundary. Solid slabs keep their thickness.

Usage:

1. Select the floor surfaces or solids
2. Enter the offset distance; a negative value pulls the edge inward

Access: Left Click



PushGlassIn

Tooltip: Creates recessed glass surfaces from selected surfaces.

Generates inset glass surfaces with surrounding frame geometry.
Useful for creating window/curtainwall details with depth.

Access: Left Click



QuickBakeTextDots

Tooltip: Quick-bake selected objects with text-dot labels.

Key Features:

- One-click bake of the current selection
- Text dots carry object names at bbox centers
- Dedicated quick-bake layer, no dialog digging
- Remembers the target layer between runs

Access: Left Click



ReplaceInPlace

Tooltip: Replace-in-place: swap same-named objects instead of duplicating.

Key Features:

- Matches incoming objects to existing by name
- Shows diff count before committing
- Deletes replaced originals, keeps newcomers
- Untouched objects and layers are never modified

Access: Left Click



ShapeMapper

Tooltip: Maps complex designs over target surfaces.

Advanced surface mapping utility that provides enhanced control compared to FlowAlongSurface.
Allows mapping of curves, surfaces and polysurfaces while maintaining design intent.

Access: Left Click



SrfToPanel

Tooltip: Advanced surface panelization utility.

Features:

- Converts surfaces to detailed panel geometry
- Configurable panel thickness and joint reveals
- Automatic edge detail generation
- Maintains design intent while adding construction detail

Access: Left Click



TimeTravel

Tooltip: Selective undo tool for Rhino objects.

Features:

- Undo history for selected objects only
- Maintains other objects' current state
- Precise history control for specific elements

Access: Left Click



ARVROverlay

Tooltip: EnneadTab-ARVR: Zero-Install Mobile Camera AR Overlay for 3D Models.

Beam your 3D models onto your smartphone camera in augmented reality:

- Export selected Rhino objects to .GLB
- Stage model in local temp folder
- Direct upload to cloud room session
- Pick existing local .GLB / .GLTF / .USDZ file to beam
- Launch Web Hub (<https://enneadtab.com/arvr>)

Access: Left Click



AiRenderUpscale

Tooltip: Upscale tool is being rebuilt - see in-app message for status.

Access: Right Click



AiRenderingFromView

Tooltip: View rendering for Rhino through Ennead's in-house style library.

Capture the active viewport, queue prompts, and see your full cloud Gallery across every device (Revit, Rhino, mobile web, desktop web - same items).

All features call the live ennead-ai.com web API. No local fallbacks: when the web product improves, this dialog improves automatically.

Access: Left Click



AssignEmptyMaterial

Tooltip: Assigns unique materials to layers without assigned materials.

Key Features:

- Creates a unique material for each layer without materials
- Material names based on layer hierarchy (including parent layers)
- Makes D5 material editing workflow more efficient
- Use the layer color as the material diffuse color

Access: Left Click



D5AssetLocator

Tooltip: Open the Render Asset Editor - the EnneadTab app for D5 and Enscape assets.

This tool has moved out of EnneadTab into its own standalone, auto-updating application, so it can ship fixes without waiting on an EnneadTab release. It now covers D5 AND Enscape in one place. This button opens its page in your browser; install it once and it keeps itself up to date from then on.

What it does:

- Locate hidden D5 asset folders across your system
- Access and modify materials on D5 objects
- Customize properties of those beautiful D5 trees, furniture and people
- Save hours of searching through obscure file directories

<https://enneadtab.com/render-asset-editor>

Access: Left Click



EnneaDuckRenderStudio

Tooltip: Open e.AI — EnneaDuck's rendering studio.

AI-powered rendering and video generation for architectural visualization:

- Multi-style image generation (Gemini + Imagen)
- Image editing and style transfer
- Video generation from stills
- Iterative conversational refinement

Opens ennead-ai.com in your default browser.

Access: Left Click



EnscapeAssetLocator

Tooltip: Open the Render Asset Editor - the EnneadTab app for Enscape and D5 assets.

This tool has moved out of EnneadTab into its own standalone, auto-updating application, so it can ship fixes without waiting on an EnneadTab release. It now covers Enscape AND D5 in one place. This button opens its page in your browser; install it once and it keeps itself up to date from then on.

What it does:

- Locate hidden Enscape asset folders across your system
- Access and modify materials on Enscape objects
- Customize properties of those beautiful Enscape trees, furniture and people
- Save hours of searching through obscure file directories

<https://enneadtab.com/render-asset-editor>

Access: Left Click



FlyThrough

Tooltip: Precise fly-through content pipeline (senzhang-todo #452).

Locks a camera path in the LIVE Rhino 3D viewport (operator-controlled, not AI-guessed), exports keyframe stills for debug/approval, then renders a final image set + optional video. Progress and failures land in Dump/FlyThrough/monitor.log + job_status.json.

How to run

1. Frame the start pose in a Perspective viewport.
2. Click this button -> Lock keyframe. Orbit/walk, lock again. Repeat. (Or Import named views to reuse cameras you already saved.)
3. Export keyframe stills and review before spending time on a full render.
4. Render final video + images. Opens the job folder when done.

Alias: FlyThrough / EA_FlyThrough

Note: This is EnneadTab-OS Rhino tooling. It is NOT the RenderPolisher AI-from-stills / Veo path (deliberately repointed off RenderPolisher).

Access: Left Click

GetEarth

Tooltip: Pull a real-world site context model into Rhino from a coordinate.

Paste a Google Maps link (or a "lat, lon" pair), give it a size, and the EnneadTab EarthModel service returns a georeferenced textured mesh you can render against. No Blender detour, no plugin to install, no API key of your own.

How to use:

1. Paste a Google Maps URL, or type "40.7128, -74.0060".
2. Give a square size in metres.
3. The model arrives on its own layer, anchored to the file's EarthAnchorPoint.

Notes:

- Good enough to RENDER. Not survey geometry, and deliberately so: do not snap, section, or measure against it.
- Google attribution is attached to the layer and must not be stripped.
- Right-click this button for the diagnostic (is the service up, am I signed in, what is cached) and for cache purge.
- The boundary-curve sizing mode described in the service plan is NOT built yet; only the square-from-a-point mode ships here. get_earth_utility already has the bbox_from_points math it will need.
- Not to be confused with GoogleEarthTutorial in the Render tab. That one is the older MANUAL route: it opens a Blender walkthrough video. This button is the automatic one. They are separate tools, not two versions of the same thing.

Access: Left Click



GetEarthSettings

Tooltip: Diagnose and configure GetEarth.

Run this first when GetEarth misbehaves. It answers the four questions that actually explain a failure:

- Can this machine reach the EarthModel service?
- Am I signed in to EnneadTab?
- What is cached locally, and how big has it got?
- Which contract version is this client speaking?

Also sets the default site size and purges the local cache.

This diagnostic SHIPS on purpose. DarkSide/ is stripped at publish, so the test harness, the stub server and every developer driver are gone on a designer's machine. Without this, a failing GetEarth button has nothing to run at all.

Access: Right Click



GoogleEarthTutorial

Tooltip: MANUAL route to site context: opens the Blender video tutorial.

This is the OLD, hand-driven way. It opens a YouTube walkthrough for capturing a city in Blender and bringing it across into Rhino, plus a companion Blender script in this button's folder that tidies up the imported materials.

There is now an AUTOMATIC one: GetEarth, in the Create tab. Paste a Google Maps link, give it a size, and the site model arrives georeferenced and textured with no Blender step at all. Try that first.

This button stays for two reasons, and will be retired when neither holds:

1. GetEarth's server-side model builder is not finished yet, so this is still the only route that works end to end today.
2. The Blender path gives you manual control over exactly what gets captured, which the automatic one deliberately does not.

Access: Left Click



ImportSelectedCamera

Tooltip: Import selected camera from another file.

Access: Left Click



InspectEnscapeSetting

Tooltip: Inspect and compare Enscape setting files for differences

Access: Left Click



LoadEnscapeToPsd

Tooltip: Load related Enscape image to Photoshop

Access: Left Click



MakeCrvPipe

Tooltip: Draw thin pipes along the curves on your [EDGE] layers so Enscape renders the edge.

Enscape will not show a line where two coplanar faces meet, which flattens joints and reveals seams that should read as crisp. A slim pipe on the curve gives it something solid to shade.

Usage:

1. Put the curves you want to read as edges on a layer with [EDGE] in its name
2. Run the button; existing pipes are rebuilt rather than duplicated

Access: Left Click



MaterialShop

Tooltip: Open AmbientCG in your browser, a free library of rendering materials.

Hundreds of ready-to-use textures, HDRI skies and props, all free to download and drop into a Rhino or Enscape scene. Good first stop when a render needs a convincing surface.

Access: Left Click



RenameEnscapeFiles

Tooltip: Rename the output of Enscape files to remove the long bit.

Access: Left Click



Block2Family

Tooltip: Turn Rhino blocks into Revit families and place them in the project.

Unlike the standard Rhino-to-Revit route, you never have to build or manage the families yourself, and a whole selection of blocks can cross over in one batch. Each block definition becomes one family, so repeated instances stay lightweight in Revit.

Usage:

1. Select the blocks you want to send over
2. Run the button and let it batch the conversion
3. Pick up the families on the Revit side

Access: Left Click



BrepToMass

Tooltip: Using faces of the brep to recreate a freeform mass in Revit.

Access: Left Click



DraftInsulationBatting

Tooltip: Draw the zig-zag insulation batting graphic along any curve you give it.

Unlike a straight hatch, the batting follows a curved or irregular path, so a detail can show insulation wrapping a corner or a rounded edge and still read correctly.

Usage:

1. Select the base curves the batting should follow
2. Enter the batting thickness

Access: Left Click



ExportCameraToRevit

Tooltip: You can recreate same 3D camera in Revit by exporting cameras from Rhino here first.

Access: Left Click



ExportMaterialByLayer

Tooltip: Export material definitions for each layer using legal file names as dictionary keys.

Access: Left Click



FloorDrafter

Tooltip: Convert brep to floor data so in Revit it can be used as floor creation base.

Access: Left Click



ImportRevitCollection

Tooltip: Organize dwgs export from Revit to readable Rhino layer tree.

Access: Left Click



LiveSelection

Tooltip: Mirror your Rhino selection over to Revit so both models highlight the same thing.

Saves hunting for the Revit counterpart of an object by hand when you are coordinating between the two, which is where most cross-platform mistakes creep in.

Access: Left Click



MapBlockTransform

Tooltip: Send the position and orientation of your Rhino blocks over to Revit.

Revit families cannot normally be tilted freely in three dimensions. Handing the block transforms to an adaptive family on the Revit side is the one route that gets true free 3D orientation, so a panel or louvre can sit at any angle it does in Rhino.

Usage:

1. Select the blocks whose placement you want to carry over
2. Run the button, then pick the transforms up from the Revit side

Access: Left Click



MapRevitSubCategoryMaterial

Tooltip: Give your Rhino layers the materials that the matching Revit subcategories use.

Export the subcategory material table from Revit first, then run this. Any layer whose name appears in that table gets the corresponding material created and assigned, so a model brought over from Revit shades the way it does there instead of arriving flat and grey.

Usage:

1. On the Revit side, run Export SubCategory Material Table
2. Run this button; layers that match a name in the table are mapped

Access: Left Click



Revit2RhinoImport

Tooltip: Brings the geometry exported by the Revit2Rhino button into this Rhino file.

Finds the latest export by itself, so there is no file to browse for.

The new objects are grouped under one layer and selected so you can move them right away.

Features:

- Imports the newest export made by the Revit2Rhino button
- Asks before importing an export that is more than half a day old
- Warns when blocks with the same names already exist in this file

Usage:

1. In Revit, run Revit2Rhino on the view you want
2. Come here and run this command

Access: Left Click



RevitDrafterExport

Tooltip: Exports drafting content to Revit.

Features:

- Processes curves and filled regions
- Maintains layer organization
- Supports multiple geometry types
- Preserves object properties

Usage:

1. Create content in OUT layers
2. Run command to export
3. Import in Revit using companion tool

Access: Right Click



RevitDrafterImport

Tooltip: Imports drafting content from Revit.

Features:

- Sets up layer structure for line styles
- Supports filled region types
- Maintains layer organization
- Preserves object properties

Usage:

1. Export content from Revit
2. Run command to import
3. Content organized in layer tree

Access: Left Click



Rhino2Revit

Tooltip: Export Layer Contents to 3dm and dwg for Rhino2Revit in EnneadTab for Revit.

Access: Left Click



Shape2Revit

Tooltip: Send loose Rhino geometry to Revit, turning each object into its own family.

For one-off shapes that were never organised into blocks. Every selected surface, solid or mesh becomes a separate Revit family, so use it sparingly or the Revit file ends up carrying hundreds of them. When your geometry is already blocked, reach for Block2Family instead: it reuses one family per definition and keeps the Revit model far lighter.

Features:

- Works with surfaces, polysurfaces and meshes
- Original shape and placement are preserved
- Your Rhino model is left exactly as you had it

Access: Left Click



SurfaceToAdaptiveComponent

Tooltip: Use the corners of the input surfs as the marker for the adaptive pts in Revit.

Access: Left Click



LibraryAttrFilter

Tooltip: Select model objects by Elefront-style key/value user-text attributes.

Rhino port of the Grasshopper library attribute filtering (EnneadTab-For-Grasshopper PR #39): pick user-text key/value criteria and every model object whose attributes match ALL criteria gets selected.

Key Features:

- Key picker lists every user-text key already in the document
- '(key present)' matches any value for a key (Elefront-style presence check)
- Stack multiple criteria (AND); values compare case-insensitively
- Reports how many objects were selected

Access: Left Click



PickSelectionContract

Tooltip: Pick-selection contract: capture viewport picks as GUIDs under a name.

Key Features:

- Prompts viewport selection at run time
- Captures picked object GUIDs into a named contract
- Re-select a stored contract in one click
- Skips deleted or missing objects on re-select

Access: Left Click



RandomDeselect

Tooltip: Randomly deselects objects based on percentage.

Features:

- Deselects random subset of selected objects
- Percentage-based deselection (1-99%)
- Remembers last used percentage
- Maintains object relationships

Usage:

1. Select objects
2. Enter deselection percentage
3. Random subset will be deselected

Access: Left Click



RandomDeselectByDist

Tooltip: Randomly deselects blocks based on their distance from a curve.

The probability of keeping a block is proportional to its distance from the curve.
Blocks closer to the curve have higher chance of being kept.
Distance clamping is available to control the influence range.

Usage:

1. Pre-select blocks or select when prompted
2. Select a base curve as attractor
3. Adjust distance clamps in the dialog
4. Click "Select" to apply random deselection
5. Keep clicking "Select" to get different random results
6. Adjust clamp values and click "Select" again to refine results
7. Click "Close" when satisfied with the selection

Features:

- Interactive Eto form with dedicated pick buttons
- Real-time distance range calculation
- Persistent settings (remembers last clamp values)
- Dynamic visualization (circles/pipes) that updates with each selection
- Iterative workflow - keep trying until you get the desired result

Access: Left Click



RandomSelectionToGroup

Tooltip: Randomly distributes selected objects into groups.

Features:

- Creates specified number of groups
- Randomly assigns objects to groups
- Useful for applying varied materials/shading
- Maintains object relationships

Usage:

1. Select objects to group
2. Specify number of groups
3. Objects will be randomly distributed

Access: Left Click



ResolveModelBinding

Tooltip: Select model objects by layer or block-name wildcard pattern.

Features:

- Layer patterns with * wildcards and :: sublayer paths (e.g. FACADE::*)
- Block-name patterns (e.g. Panel_*)
- Remembers the last pattern used

Usage:

1. Choose layer or block-name matching
2. Enter a wildcard pattern
3. Matching objects are selected

Access: Left Click



BakeGFADataToExcel

Tooltip: Export GFA (Gross Floor Area) data to Excel and manage area targets.

Features:

- Export area calculations to formatted Excel spreadsheet
- Generate checking surfaces for visual verification
- Set and manage target areas for different GFA categories
- Compare actual vs target areas with variance analysis

Usage:

- Click to export current GFA data to Excel
- Right-click to access additional options:
- Generate checking surfaces
- Set target areas for GFA categories
- Edit existing target values

Access: Right Click



BatchExportRhinoView

Tooltip: Exports multiple Rhino views in batch.

Allows selection of multiple views for automated export.
Supports customizable resolution and format settings.

Access: Left Click



BatchRenameCamera

Tooltip: Enables bulk camera renaming operations.

Provides interface for renaming multiple cameras without activation.
Maintains camera properties while updating names efficiently.

Access: Left Click



ChangeObjectDisplaySource

Tooltip: Modifies object display source settings.

Controls whether objects inherit color and material properties from their layer.
Provides batch modification capabilities for multiple objects.

Access: Left Click



SectionBoxCleanup

Tooltip: Reset the view to unbounded.

Access: Left Click



SectionboxByBoundingBox

Tooltip: Creates section boxes around selected elements.

Similar to Revit's SectionBox, crops views to selected element boundaries.
Supports all clipping modes in Rhino, with limited Enscape compatibility.

Access: Left Click



SectionboxByPolysrf

Tooltip: Use closed polysrf as input box cutter.

Access: Right Click



ToggleGFA

Tooltip: Toggles Gross Floor Area (GFA) visualization and calculation.

Features:

- Processes layers marked with [GFA] to calculate and display area information
- Real-time updates as geometry changes
- Supports area factor multipliers using $\backslash\{factor\}$ syntax in layer names
- Excel data export capabilities
- Automatic unit conversion (mm/m -> SQM, inch/ft -> SQFT)
- Dynamic merging of coplanar surfaces at same elevation
- Support for single surfaces and polysurfaces
- Live comparison of how much is off from target.

Usage:

- Add [GFA] to layer names to include in calculation
- Optional $\backslash\{factor\}$ at end of layer name for area multipliers (e.g. $\backslash\{0.5\}$)
- Right-click to export to Excel or generate checking surfaces or set target area for each keyword.

Access: Left Click



ToggleLayerPointer

Tooltip: Displays quick layer information for visible objects.

Creates a filtered layer list showing only layers with visible objects.
Useful for examining and understanding model layer structure.

Access: Left Click



ViewBlockDiagram

Tooltip: Draw the view fan from a chosen viewpoint, showing what is visible and what is blocked.

Sight lines are traced outward from the viewer and stopped wherever a building or other obstacle gets in the way. The clear sweeps come back as surfaces you can present, so a window, a terrace or a public space can be argued for on the strength of what it actually sees.

Usage:

1. Set the sweep resolution and how far the view should reach
2. Select the objects that block the view
3. Pick the viewer point; the visible fans are drawn for you

Access: Left Click



ViewToggle

Tooltip: Flip the active viewport between Top and Perspective with one click.

Saves reaching for the viewport tabs every time you want to check a plan relationship and then get back to the model. Running it also binds the F4 key to the same toggle, so from then on you can switch with the keyboard alone.

Access: Left Click



ListenToMiro

Tooltip: Listen to changes in the miro

Access: Left Click



PushToMiro

Tooltip: Push selected elements in Rhino to Miro. Only support text and rect and circle.

Access: Right Click

Tailor Scripts Information

This documentation excludes EnneadTab Tailor scripts from the main content.

10 tailor-related scripts were skipped for **Rhino**.

Tailor scripts are project-specific customizations and are not included in the general documentation.